Brief mindfulness training and Stroop interference in university students

Chiara Verdi
Department of General Psychology, University of Padova

Marco Neri
Department of General Psychology, University of Padova

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Abstract

We tested whether four weeks of brief mindfulness training reduce Stroop interference. Forty university students were randomly assigned to a mindfulness or a control group. Interference was smaller in the mindfulness group, but the difference did not reach significance, t(36) = 1.86, p = .071, d = 0.28. Across participants, higher dispositional mindfulness was associated with smaller interference, r(36) = −.47, p = .003.

Method

Participants

Forty students of the University of Padova took part (20 per group). Two participants with accuracy below 80% were excluded, leaving 38 participants (19 per group; 22 women; mean age = 23.8 years, SD = 2.4).

Task and procedure

Participants completed a colour-word Stroop task (96 trials, half congruent) programmed in PsychoPy, and the Mindful Attention Awareness Scale (MAAS). Trials with reaction times below 200 ms or above 2000 ms were excluded. Interference was computed for each participant as the difference between the mean reaction time of correct incongruent and correct congruent trials.

Results

Mean interference was 86.7 ms (SD = 49.7) in the control group and 73.7 ms (SD = 42.5) in the mindfulness group. The difference between groups was not significant, t(36) = 1.86, p = .071, d = 0.28.

Interference correlated negatively with MAAS scores, r(36) = −.47, 95% CI [−0.68, −0.17], p = .003 (Figure 1).

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.